

Development of a Wireless Active Personal Dosimeter for Interventional Radiology

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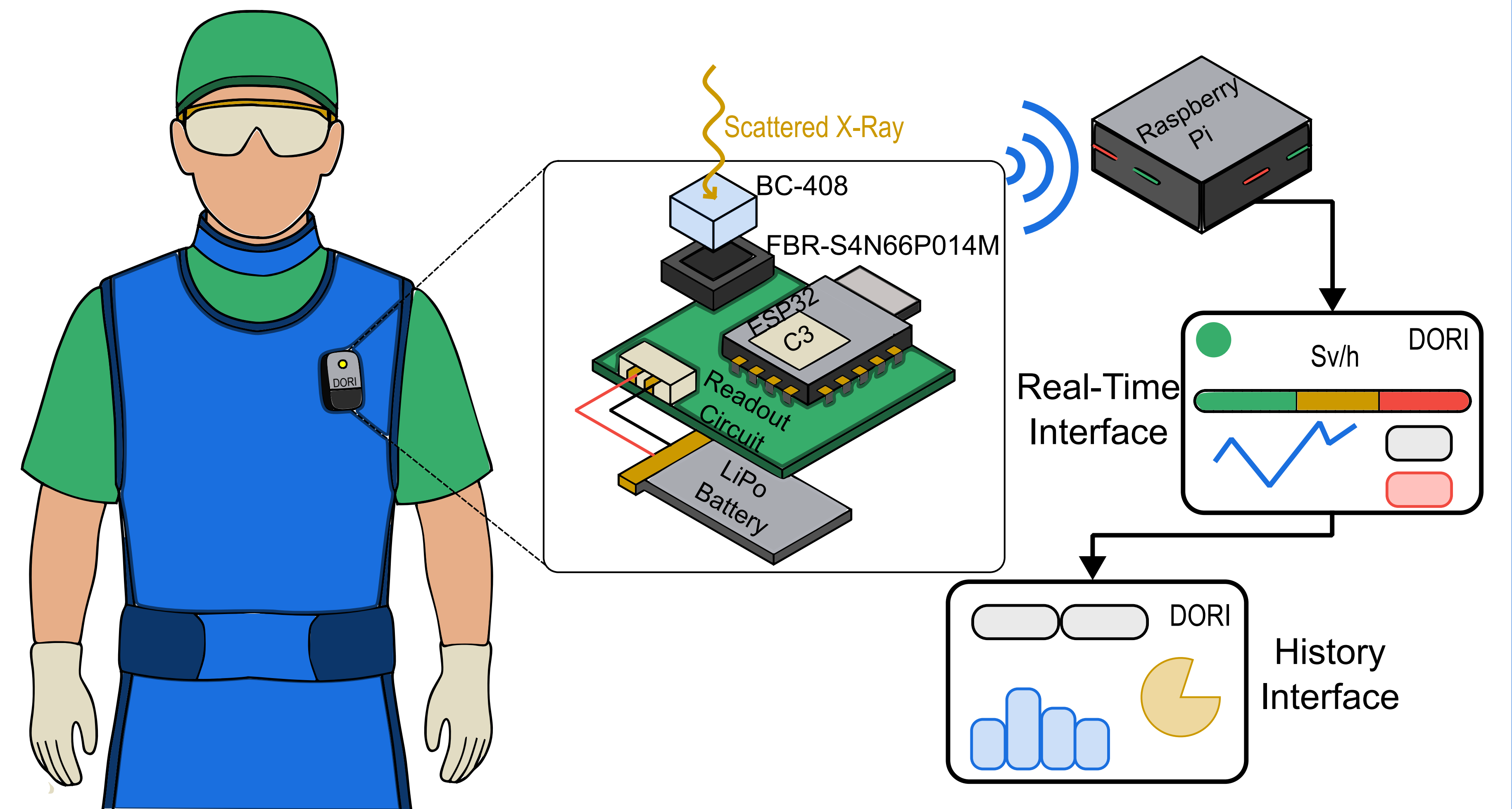
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ABSTRACT

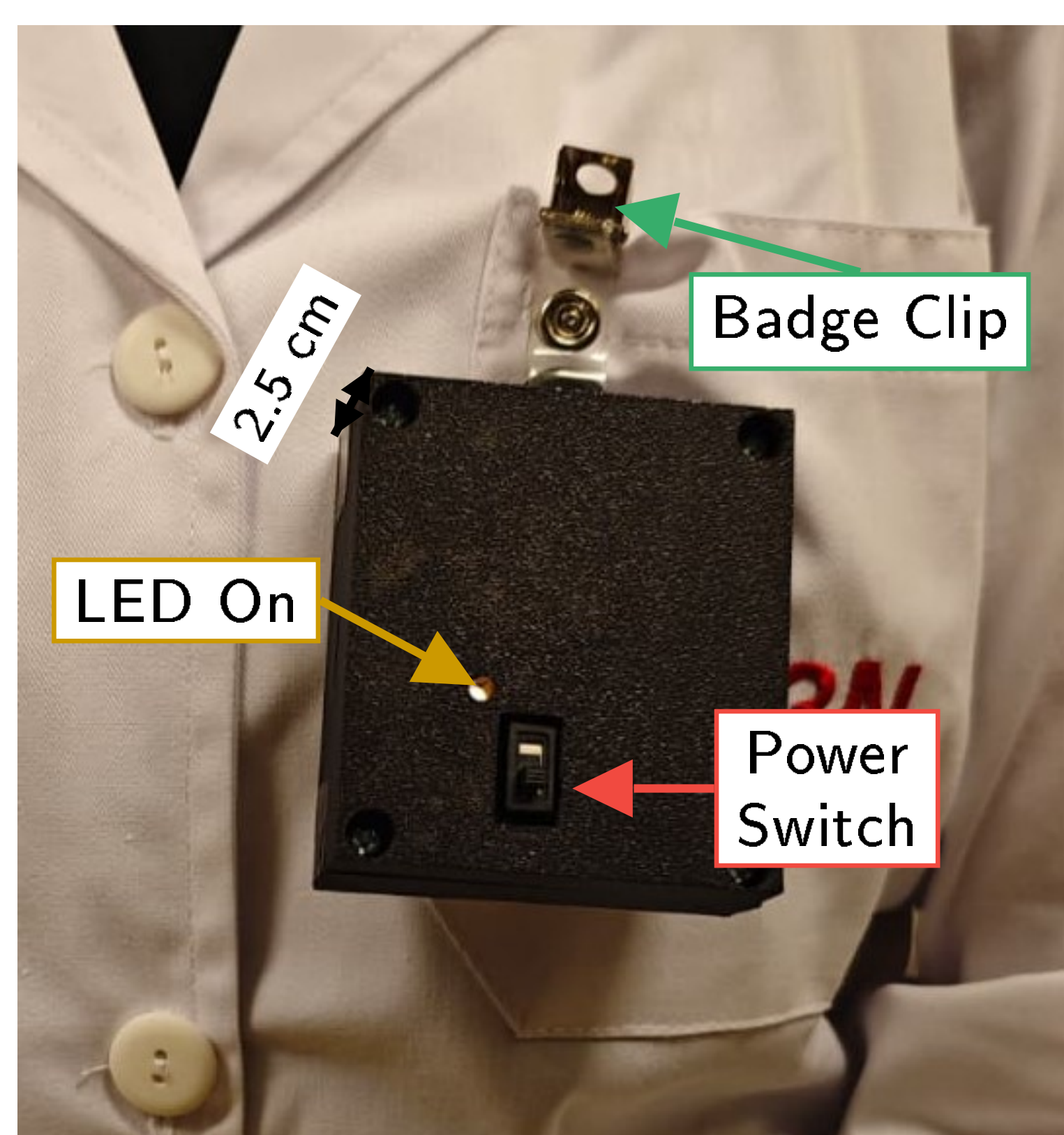
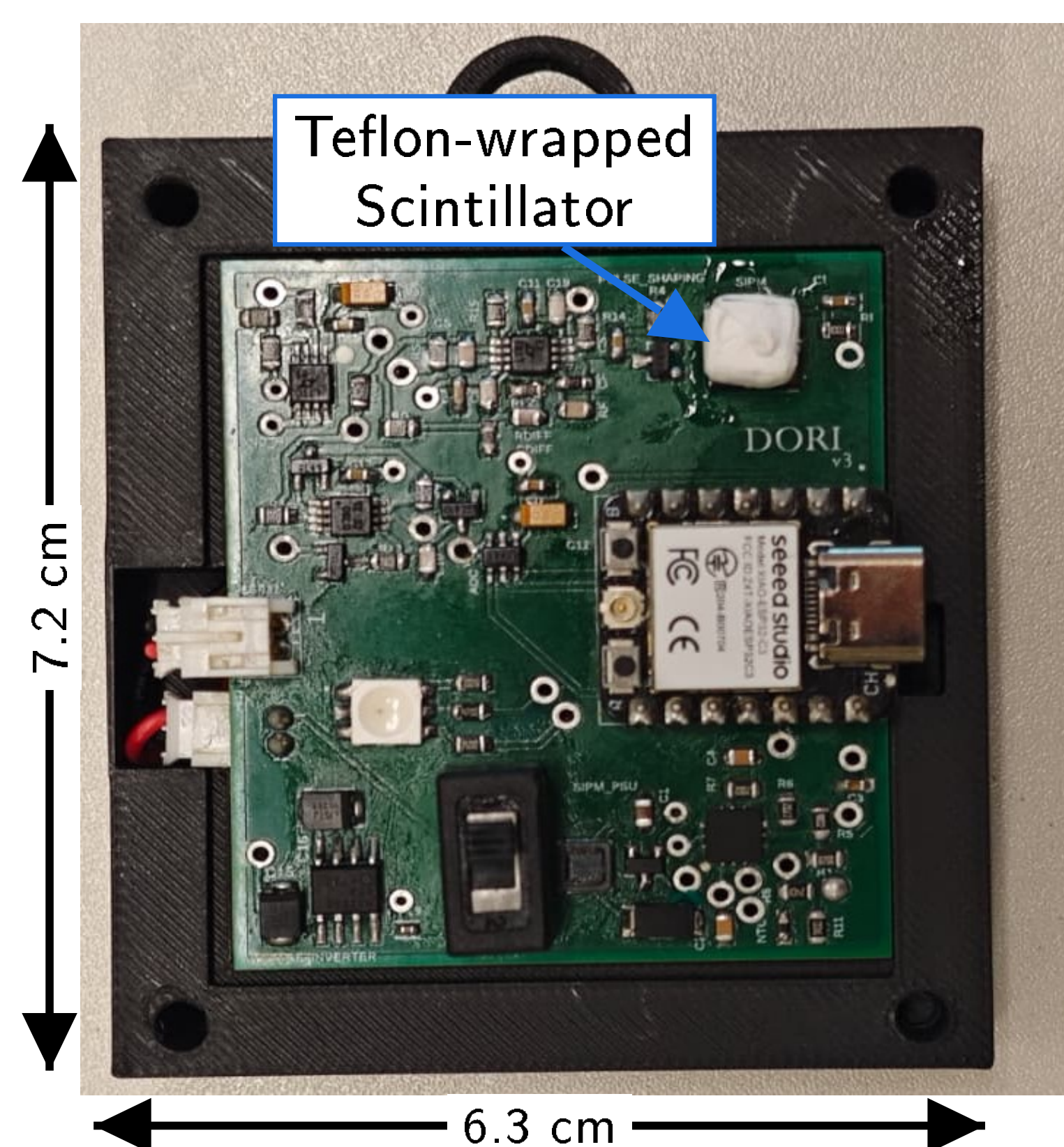
Interventional radiology is the medical specialty subject to the highest occupational exposure to ionising radiation [1]. Monitoring still relies on passive dosimeters, which report dose retrospectively, so unnecessary exposure cannot be anticipated [2].

This work presents DORI (*DO*símetro para *R*adiologia de *I*ntervenção), a wireless active personal dosimeter for real-time monitoring of scattered radiation. A plastic scintillator coupled to a silicon photomultiplier is read out by custom front-end electronics and an ESP32-C3. The energy scale was calibrated with bremsstrahlung endpoints and the ²⁴¹Am photopeak; the dose response was calibrated against ISO 4037 narrow spectra in terms of $H_p(10)$. The response is linear from 20 to 60 keV and the reconstructed $H_p(10)$ follows the reference values.

METHODS

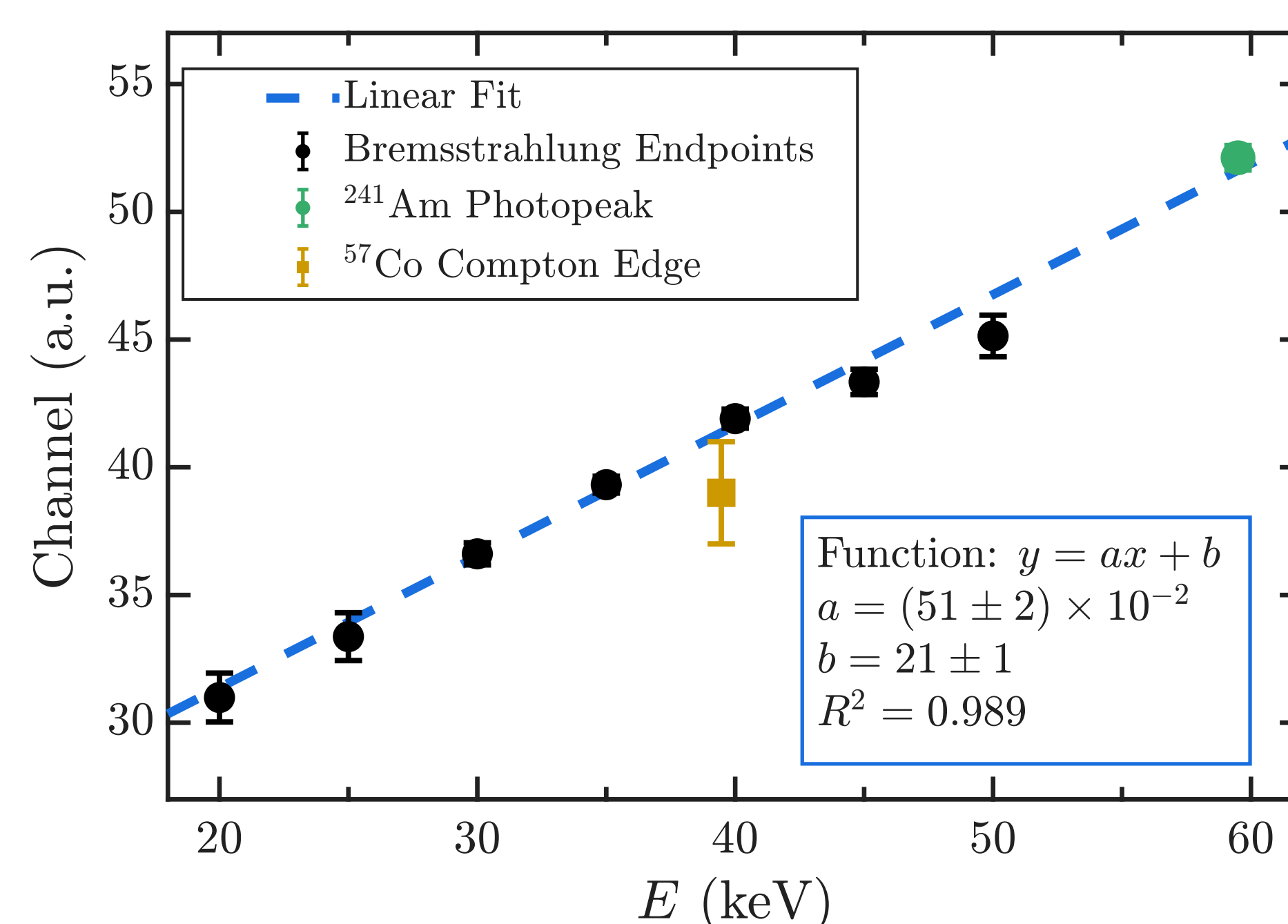


PROTOTYPE

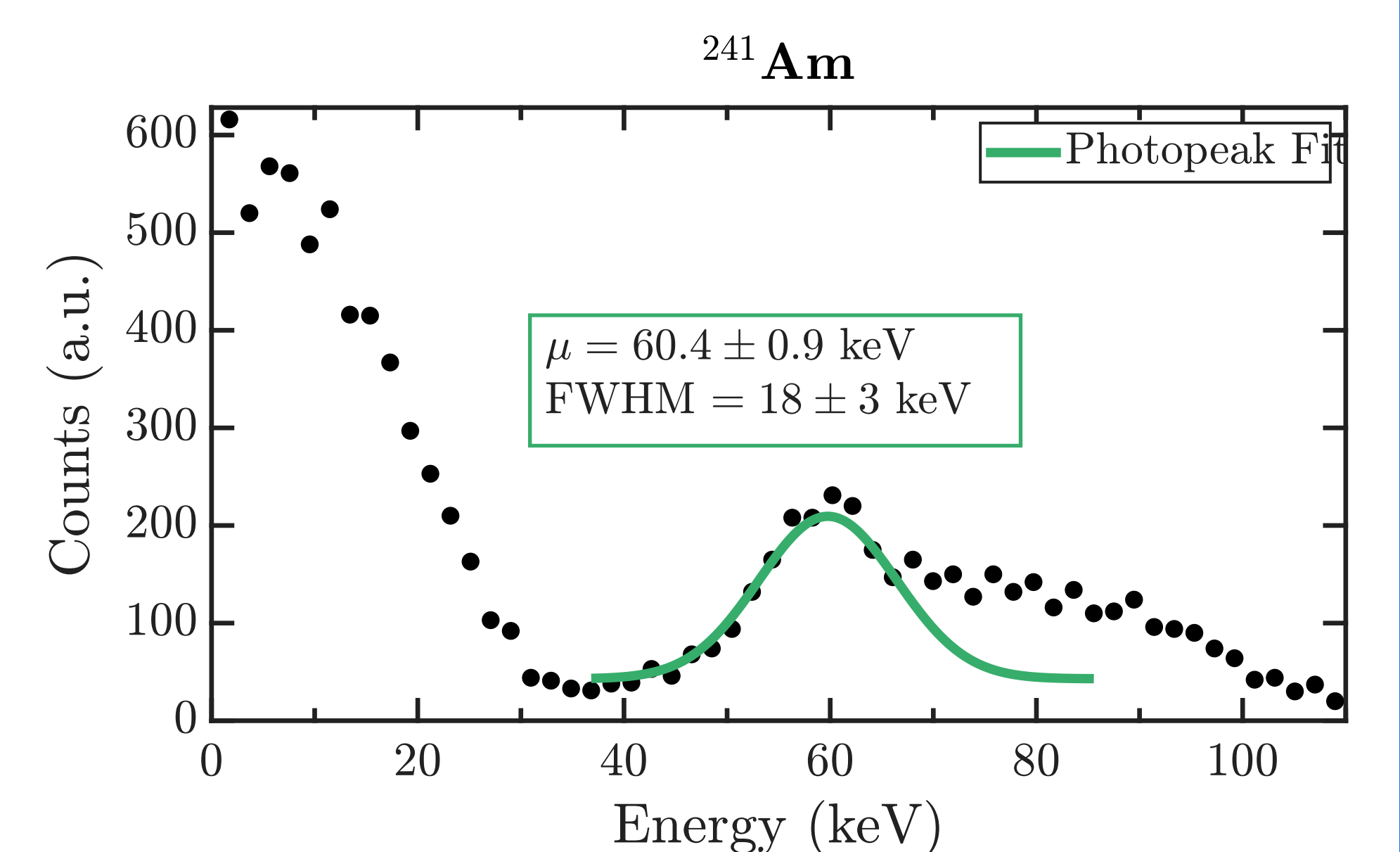


Parameter	Value
Dead time	$17.8 \pm 0.4 \mu\text{s}$
Energy range	20 to 60 keV
Operational unit	$H_p(10)$
Dose range	N-25 to N-40
Dimensions	$6.3 \times 7.2 \times 2.5 \text{ cm}$
Mass	141 g
Autonomy	8.6 h

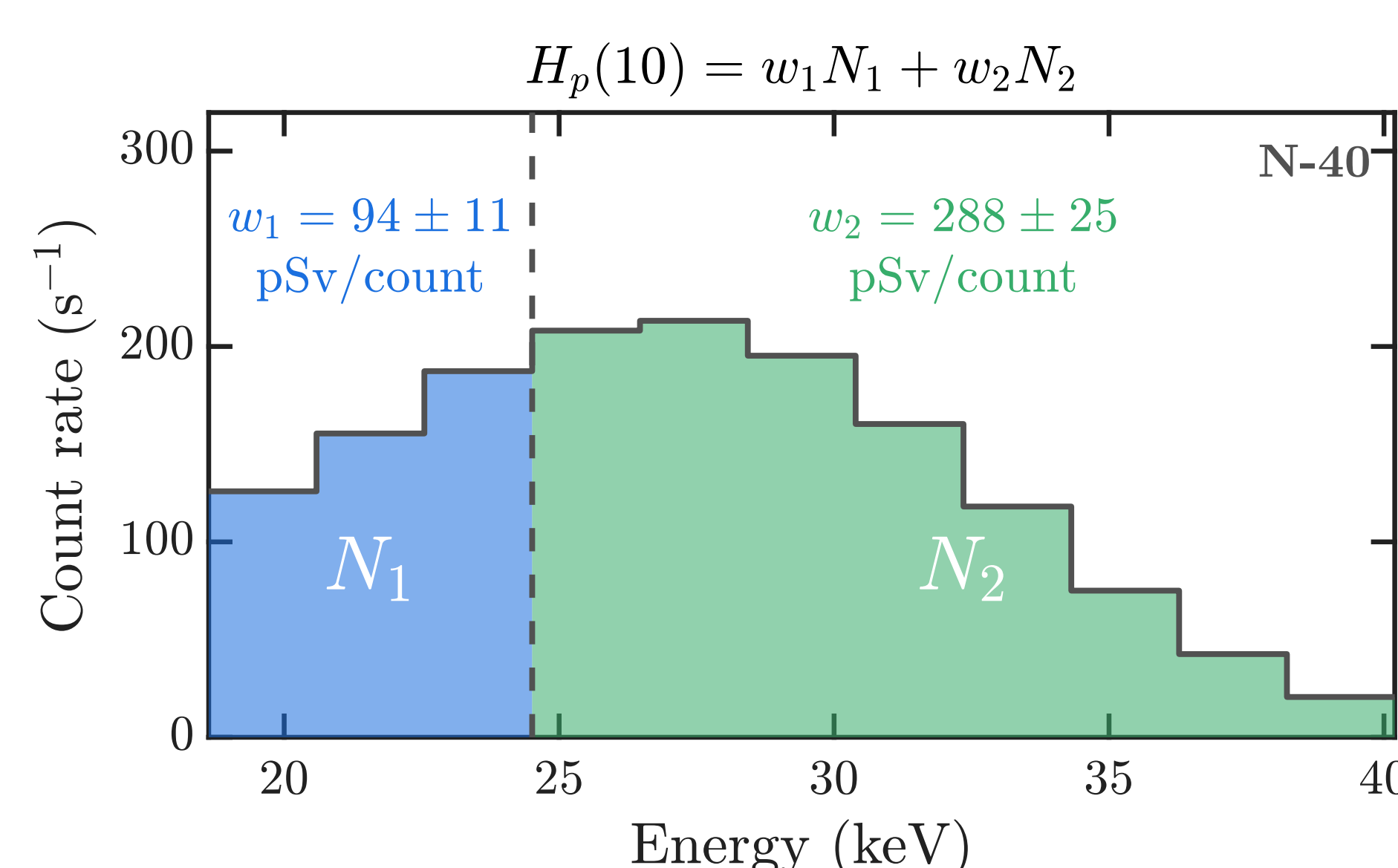
RESULTS



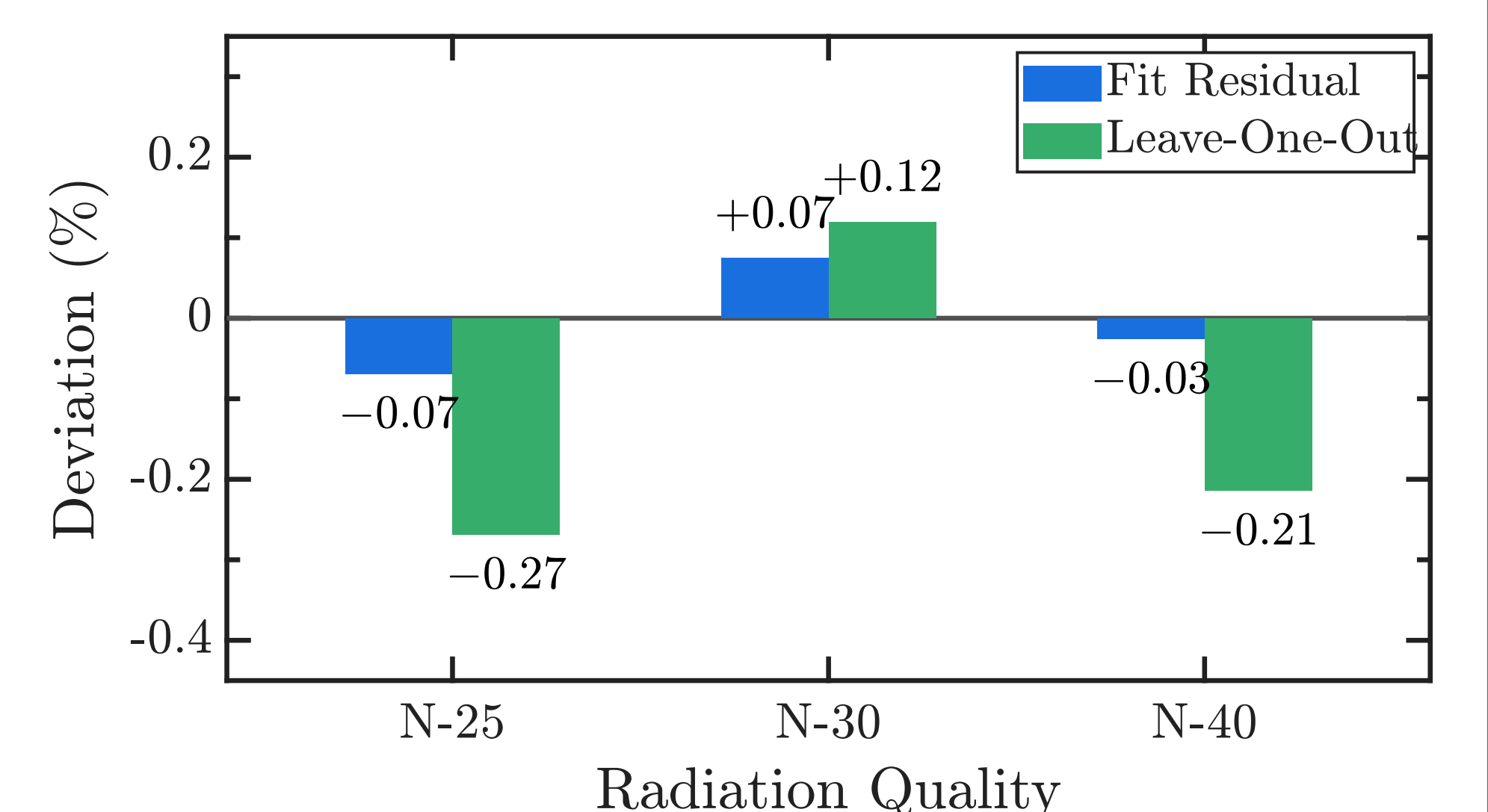
Linear response between 20 and 60 keV



Energy resolution of 30% at 59.5 keV



Dose read from two energy bins



Reconstructed $H_p(10)$ follows the reference

CONCLUSIONS

A functional wireless active personal dosimeter was built and characterised. The plastic scintillator is a feasible detector, and both calibration methods were shown to work. Future work will extend both calibrations to a wider energy range and to more reliable dose points.

REFERENCES

- [1] <https://doi.org/10.1259/bjr.20210436>
- [2] <https://doi.org/10.1007/s00330-025-11566-5>
- [3] ISO 4037-1:2019, *Radiation characteristics and production methods*.
- [4] ISO 4037-3:2019, *Calibration of area and personal dosimeters*.